MILITARY SPECIFICATION

STOPPER, BOTTLE, WHITE RUBBER

This limited coordination Military Specification has been prepared by the Defense Personnel Support Center based upon current available technical information, but it has not been approved for promulgation. It is subject to modification. However, pending its promulgation as a coordinated Military Specification, it may be used in procurement.

1. SCOPE

1.1 This specification covers white rubber bottle stoppers.

1.2 Classification. Stoppers shall be furnished in the following sizes, as specified, 3, 4, 5, 6, 7, 8, 9, and 10 (see 6.3).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

MMM-A-250	Adhesive Water-resistant (For Sealing Fiberboard Boxes).
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing).

STANDARDS

Federal

FED-STD-601	Rubber Sampling and Testing.

FSC 6640

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Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

3. REQUIREMENTS

3.1 Material. Material shall be laboratory grade, white, resilent. natural rubber.

3.2 Construction. Stoppers shall have perforations which extend from the top to the bottom of the stopper. The perforations shall be parallel to the vertical axis of the stopper and shall be symetrically located (see 3.2.1). Diameter of perforations shall be 5 ± 0.5 millimeters (mm). Perforations shall be countersunk on the top surface and shall be sealed at the bottom by nodules The nodules shall be so constructed that they can be twisted off by the application of finger pressure. Removal of nodules by twisting shall result in an unobstructed opening which permits the insertion of glass tubing.

3.2.1 Number of perforations. Sizes 3 and 4 shall have two perforations. Sizes 5, 6, 7, 8, 9 and 10 shall have three perforations.

3.3 Dimensions. Length of stoppers shall be 25 mm for all sizes. Top and bottom diameters shall be as indicated in Table I. Tolerance on length, top diameter and bottom diameter shall be ± 1 mm.

Table I. - Dimensions

Size	Top diameter mm	Bottom diameter mm
3	24	18
4	26	20
5	27	23
6	32	26
7	37	30
8	41	33
9	45	37
10	50	42

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3.4 Hardness. The stopper shall show an indentation from 36 to 54 when tested as specified in 4.4.1.

3.5 Resistance to steam. No dimension shall change by more than five percent when tested as specified in 4.4.2.

3.6 Identification marking. Each package of stoppers shall be legibly and permanently marked with the manufacturer's name or registered trademark, and shall bear the following clearly printed legend: "CAUTION - Stoppers are rubber based and may become tacky when used with organic solvents." Each stopper shall bear the applicable stopper number. Stopper number shll be raised.

3.7 Workmanship. Workmanship shall be first class throughout. The stopper shall be free from defects listed in table III and other defects which detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels, acceptable quality levels and units of product shall be as indicated in table II.

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Table II. - Sampling for examination

Type defect	Unit of Product	Inspection level	AQL (percent defective)
For visual examination			
Major A	Stopper	II	1.0
Major B	Package	II	1.0
Minor A	Stopper	II	4.0
Minor B	Package	II	4.0
For dimensional examination	Stopper	S-3	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105. Units of product shall be the stopper and acceptance number shall be zero. Inspection level shall be as indicated in table III.

Table III. - Sampling for tests

Characteristic	Requirements	Test procedure	Inspection level
Hardness	3.4	4.4.1	S-3
Resistance to Steam	3.5	4.4.2	S-3

4.3 Examination. The stoppers and immediate containers shall be examined to determine compliance with all requirements contained in this specification. The stoppers and immediate containers shall be examined for defects including, but not limited to, those listed in table IV.

Table IV - Classification of defects

Categories	Defects
Major A:	
101	Stopper not symmetrical or free of deformation.
102	Stopper not free of blooming, bleeding, brittleness or tackiness.
103	Stopper not free of cuts, cracks, blisters or ingrained particles.
104	Stopper not free of foreign matter.
105	Size identification marking not present, complete, correct or legible.

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Table IV - Classification of defects (Con't)

Major B:	
111	Caution notice not present, complete, correct or legible.
112	Immediate container not sealed or free of tears.
Minor A:	
201	Surface of stopper not smooth (other than raised marking)
202	Edges not clean cut.
Minor B:	
211	Manufacturer's name or trademark not present, complete, correct or legible.

4.3.1 Dimensional examination. The stoppers shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Hardness test. The stopper shall be tested for hardness in accordance with Method 3021 of Federal Test Method Standard No. 601.

4.4.2 Test for resistance to steam. Top outside diameter, bottom outside diameter and length shall be measured and the measurements recorded. Stopper shall be suspended in an atmosphere of steam in an autoclave for 120 hours. Exhaust tube of the autoclave shall be opened sufficiently to keep the temperature of the steam at 100 $\pm$ 3° Centigrade and steam shall be constantly flowing out of the exhaust tube. After the 120 hours in the autoclave stopper shall be kept at room temperature for 24 hours. Top and bottom diameters and length shall be measured and percentages of change calculated.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One package (number of stoppers), as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in the following table shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

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Table V. - Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	10 units	40 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptance to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each unit of issue shall be packaged in a sealed clear plastic bag.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2 inch wide tape conforming to PPP-T-76. Closing of bottom flap with adhesive conforming to MMM-A-250 or metal stitches is permitted.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

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5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown. In addition, marking specified in 3.6 shall be shown.

5.3.2 Intermediate package. Each intermediate package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. For general use in the laboratory as solid stoppers and as perforated stoppers.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packing (see 5.1 and 5.2).

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6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the requirements for the following Federal Stock Numbers:

6640-935-1102, Stopper, Bottle, White Rubber, Solid or Two Hole, No. 3, 50s.

6640-935-1103, Stopper, Bottle, White Rubber, Solid or Two Hole, No. 4, 50s.

6640-935-1104, Stopper, Bottle, White Rubber, Solid or Three Hole, No. 5, 40s.

6640-935-1105, Stopper, Bottle, White Rubber, Solid or Three Hole, No. 6, 30s.

6640-935-1106, Stopper, Bottle, White Rubber, Solid or Three Hole, No. 7, 20s.

6640-935-1107, Stopper, Bottle, White Rubber, Solid or Three Hole, No. 8, 20s.

6640-935-1108, Stopper, Bottle, White Rubber, Solid or Three Hole, No. 9, 15s.

6640-935-1109, Stopper, Bottle, White Rubber, Solid or Three Hole, No. 10, 10s.

6640-935-1122, Stopper, Bottle, White Rubber, Assorted, 9 No. 3s, 8 No. 4s, 6 No. 5s, 4 No. 6s, 2 No. 7s, 2 No. 8s, 1 No. 9, 1 No. 10.

6.4 Specified sizes correspond to commercial sizes. Commercial sizes 1 and 2 are not covered by this specification. In case of conflict between specified and commercial sizes the requirements of this specification shall rule.

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements.

SPECIFICATION

MIL-S-36662(DSA-DM) FSN:

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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